VibeFinder 2.0: An Explainable AI Music Recommendation System

Natural-language music requests → retrieval → ranking → explanation → reliability checks

Request > Retrieve > Rank > Explain > Verify

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Extending a Transparent Recommender

Base Project → Extended AI System

Original Project

- Transparent content-based music recommender
- Structured inputs: Genre, Mood, Energy, Acoustic preference
- Ranked songs with explicit scoring reasons
- **Limitation: Not conversational or context-aware**

New AI Extensions

- Natural-language query parsing & intent extraction
- Multi-source retrieval (Knowledge Base + Support Docs)
- Agent-style intermediate trace for transparency
- Reliability suite and evaluation harness

AI-NATIVE EXTENSION

System Architecture

A Robust Pipeline for Grounded Recommendations

Parser

Extracts intent, genre, mood, and activity tags from natural language.

Retriever

Checks song knowledge base and query support documents.

Recommender

Scores songs using metadata and retrieval bonuses.

Reliability

Reruns pipeline to check consistency and ground results.

System Demo

From Natural Language to Grounded Results

NORMAL CASE

> *"Need calm music for late-night coding"*

- Parser: focus/late-night intent
- Retrieval: focus-oriented evidence
- Midnight Coding, Focus Flow

SPECIALIZED STYLE

> *"Need high-energy music for the gym"*

- Style: Hype Trainer
- Boosted workout interpretation
- Gym Hero, Storm Runner

EDGE CASE

> *"Need classical music for an intense workout"*

- Surfaces intent tradeoffs
- Genre safeguard preserved

Reliability Warning: Thin catalog support for contradictory request.

Key Applied AI Features

Enhancing Transparency and Reliability

Multi-source Retrieval

Uses **Knowledge Base** and **Support Docs** to interpret synonym-heavy prompts like "spin class intervals".

Agent-style Workflow

Intermediate steps are fully visible: **parse, retrieve, plan, rank, and self-check** for transparency.

Specialized Behavior

Style-card-based explanations tailored to user context.

focus_coach **hype_trainer** **reflective_curator**

Reliability & Evaluation

Consistency reruns, warning systems, and a full **evaluation harness** to measure performance.

Measurable Improvements

How I Measured Whether It Works

18 / 18
Functional Tests Passed

4 / 4
Reliability Benchmarks

0.95
Average Confidence Score

Metric	Baseline	Enhanced
Hit@3 Accuracy	5 / 7	**7 / 7**
Trace Visibility	None	**7 Steps Avg**
Style Compliance	N/A	**100%**
Retrieval Sources	1 Source	**Multi-Source**

Building Trust Through Explainable AI

What I Learned as an AI Engineer

Key Engineering Lessons

AI systems need more than "reasonable output"; they require **reliability and transparency** to be useful.

- Retrieval quality is a primary driver of performance, often matching the importance of ranking logic.
- Building guardrails and reliability checks is a core part of AI engineering, not an optional extra.

REFLECTION

"I approach AI engineering as systems work. I care about building tools that are explainable, testable, and honest about their limits."

VibeFinder 2.0 demonstrates practical systems that are